Multimodal MRI Fusion and Missing-Modality Learning for BraTS Glioma Segmentation: A Comprehensive Survey and Taxonomy

ABSTRACT

Glioma segmentation from multiparametric MRI relies on four standard BraTS sequences—T1, T1ce, T2, and FLAIR—yet clinical deployment is hindered by missing modalities due to artifacts, contrast-agent contraindications, or protocol variation. This survey provides a systematic, evidence-based comparison of multimodal fusion strategies and missing-modality learning methods across 31 primary studies spanning BraTS 2012–2025. We develop a six-axis, multi-label taxonomy (fusion strategy, encoder design, fusion location, missing-modality strategy, availability awareness, evaluation coverage) and populate four analytical benchmark matrices—segmentation performance, robustness to missing modalities, computational efficiency, and clinical reliability—exclusively with explicitly reported numerical results, preserving BraTS edition, dataset split, metric scale, and missing-modality configuration. Key findings include consistent modality-contribution asymmetry (FLAIR dominant for whole tumor, T1ce for tumor core and enhancing tumor), superior robustness of hybrid and adaptive fusion with knowledge distillation over synthesis-based approaches, and a near-total absence of uncertainty quantification, calibration, or external clinical validation in the reviewed corpus. Seven critical research gaps are identified with concrete future directions, including a standardized missing-modality benchmark, uncertainty-aware evaluation, and computational–clinical Pareto reporting. This work serves as a critical reference for method selection and research planning in medical image analysis.

Keywords: multimodal MRI fusion, missing-modality learning, BraTS glioma segmentation, knowledge distillation, clinical reliability

1. Introduction

1.1 Motivation & Background

Glioma segmentation from multiparametric MRI (mpMRI) is essential for diagnosis, treatment planning, and response assessment [4], [11]. The Brain Tumor Segmentation (BraTS) challenge has standardized four complementary sequences—T1-weighted (T1), contrast-enhanced T1 (T1ce), T2-weighted (T2), and Fluid-Attenuated Inversion Recovery (FLAIR)—and three tumor subregions: enhancing tumor (ET), tumor core (TC), and whole tumor (WT) [11], [15]. Despite remarkable progress on full-modality data, with nnU-Net achieving 88.95% WT, 85.06% TC, and 82.03% ET Dice on the BraTS 2020 challenge test set [11], clinical deployment faces a fundamental obstacle: **missing modalities** due to motion artifacts, contrast-agent allergy, protocol variation, or resource constraints [4], [7], [9], [19], [24]. Missing FLAIR causes sharp Dice degradation for WT; missing T1ce severely impairs TC and ET; missing T1 or T2 yields milder drops [2]. This asymmetry reflects the distinct pathophysiological information each modality carries and has been consistently observed across independent studies [2], [27].

The literature has responded with diverse fusion strategies—early concatenation, late ensembling, hybrid shared/specific encoders, attention-based gating, modality-specific pathways, and adaptive dynamic routing—and missing-modality learning families including modality dropout, modality-specific encoding, shared latent representations, knowledge distillation, modality synthesis/reconstruction, and adaptive missing-modality-aware fusion. However, these methods have been evaluated under heterogeneous protocols: BraTS editions 2012–2025, validation versus hidden-test splits, Dice reported as 0–1 or percentage, predefined single-modality dropout versus 15 combinatorial configurations [3] versus arbitrary subset evaluation [9], [18], [19], [23], and almost no reporting of computational cost or clinical reliability indicators. This heterogeneity prevents fair cross-study comparison and obscures which design choices genuinely improve robustness versus those that overfit to specific evaluation settings.

1.2 Limitations of Existing Surveys

Prior reviews in the retrieved corpus either focus on full-modality segmentation [12] or treat missing-modality learning as a monolithic category without systematic contrast against fusion strategy. No survey in the corpus has: (i) organized methods along orthogonal axes that separate fusion mechanism from missing-modality handling and availability awareness; (ii) distinguished predefined missing-modality configurations from arbitrary subset evaluation; (iii) separated validation-set from hidden-test results, or full-modality from missing-modality performance; (iv) mapped computational efficiency and clinical reliability metrics alongside segmentation accuracy. The coverage of earlier surveys was not systematically compared in the retrieved passages; therefore, we do not claim to supersede them but rather to fill the specific evidence-synthesis gaps listed above.

1.3 Research Gaps and Research Questions

The evidence synthesis reveals seven critical gaps: absence of a standardized missing-modality benchmark, no uncertainty quantification or calibration reporting, limited external clinical validation beyond curated BraTS challenge test sets, missing computational–clinical Pareto data, unresolved synthesis-versus-distillation trade-off for segmentation, no systematic comparison of teacher/distillation strategies, and no failure-mode taxonomy for confounders such as resection cavities or non-GBM histologies. These gaps motivate the following research questions:

RQ1. How do existing multimodal fusion strategies (early, late, hybrid, attention-based, modality-specific encoder, adaptive) and missing-modality learning families (dropout, specific encoding, shared latent, distillation, synthesis, adaptive-aware) compare on BraTS glioma segmentation across the four analytical dimensions — segmentation performance, robustness to missing modalities, computational efficiency, and clinical reliability — when evaluated on explicitly reported numerical evidence?

RQ2. What are the methodological limitations and evaluation-protocol heterogeneities (BraTS edition mismatch, split inconsistency, missing-modality configuration non-standardization, metric scale variance, absent uncertainty/calibration/failure-detection reporting, computational transparency gap) that prevent fair cross-study comparison and clinical translation?

RQ3. Which evidence-supported research gaps (standardized missing-modality benchmark, uncertainty quantification and calibration, external clinical validation, computational–clinical Pareto frontier, modality synthesis utility for segmentation, teacher/distillation strategy optimization, failure mode taxonomy) are most critical, and what concrete future directions does the synthesized evidence suggest for each?

RQ4. How does the consistent modality-contribution asymmetry (FLAIR→WT, T1ce→TC/ET, T1/T2 complementary) inform the design of availability-aware fusion, teacher election, gating mechanisms, synthesis priority, and evaluation protocols for robust missing-modality learning?

1.4 Core Contributions

- **Unified multi-axis taxonomy with multi-label codes for 31 methods** across six axes (fusion strategy, encoder design, fusion location, missing-modality strategy, availability awareness, evaluation coverage), enabling systematic cross-family comparison and revealing that hybrid and adaptive fusion combined with knowledge distillation dominate the robustness landscape.
- **Evidence-dense benchmark matrices** for four analytical dimensions populated *only* with explicitly reported numbers from 11 of 31 papers (217 verified records), preserving BraTS edition, split, metric scale, missing-modality configuration, and uncertainty markers (NR for unreported), exposing that no method reports calibration or failure detection and only one reports training time [8].
- **Critical gap analysis with seven evidence-supported research gaps**, each structured as observed finding, specific missing evidence, significance, and concrete future direction (e.g., BraTS-Missing benchmark with fixed 15/31/63 subset configurations, hidden test, and uncertainty reporting).
- **Methodological limitation audit** documenting evaluation-protocol heterogeneity (edition mismatch, split inconsistency, configuration non-standardization, metric scale variance), clinical reliability evidence void, computational transparency gap, and unresolved synthesis-versus-distillation trade-off.

1.5 Paper Organization

Section 2 presents the six-axis taxonomy and maps all 31 methods to multi-label codes. Section 3 synthesizes the empirical benchmark matrices: full-modality segmentation (Table 4), single-modality missing (Table 5), multiple-modality missing (Table 5), synthesis quality (Table 6), computational efficiency (Table 7), and clinical reliability indicators (Table 8). Section 4 analyzes technical mechanisms and trade-offs across fusion mechanisms (Table 9), missing-modality strategies (Table 10), and modality-contribution asymmetry. Section 5 discusses methodological limitations, the seven critical research gaps (Table 11), and a synthesis toward a unified three-tier architecture. Section 6 provides the quantitative evidence ledger (Table 12). Section 7 concludes with answers to the four research questions and a call for standardized, uncertainty-aware, clinically grounded evaluation.

2. Review Methodology

2.1 Research Questions

RQ1. How do existing multimodal fusion strategies (early, late, hybrid, attention-based, modality-specific encoder, adaptive) and missing-modality learning families (dropout, specific encoding, shared latent, distillation, synthesis, adaptive-aware) compare on BraTS glioma segmentation across the four analytical dimensions — segmentation performance, robustness to missing modalities, computational efficiency, and clinical reliability — when evaluated on explicitly reported numerical evidence?

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2.2 Corpus, Search Strategy and Selection Criteria

This review is a narrative synthesis of a curated internal PDF repository; it is **not** a systematic review and no PRISMA flow was applied. The repository catalog contains 50 papers. A hybrid dense+sparse retrieval pipeline (embedding model BAAI/bge-m3, reranker BAAI/bge-reranker-v2-m3, hybrid score weights dense 0.7 / sparse 0.3, at most two chunks per paper retained) was executed on 2026-11-10 with five queries:

- "BraTS glioma segmentation multimodal fusion strategies early fusion late fusion hybrid fusion attention-based fusion modality-specific encoders adaptive fusion survey review"
- "BraTS missing modality learning glioma segmentation modality dropout knowledge distillation modality synthesis reconstruction shared latent representation missing-modality-aware fusion"
- "BraTS benchmark datasets BraTS 2018 2019 2020 2021 2022 2023 2024 evaluation metrics Dice score Hausdorff distance whole tumor tumor core enhancing tumor validation test split"
- "multimodal MRI fusion missing modality robustness computational efficiency clinical reliability external validation uncertainty estimation calibration failure detection glioma segmentation"
- "comparative analysis multimodal fusion missing modality methods BraTS segmentation performance quantitative results SOTA comparison empirical evidence original studies"

Thirty-one distinct papers were returned by at least one query and constitute the evidence base of this manuscript. The remaining 19 catalog papers were not retrieved; this is a **retrieval limitation** (relevant work may have been missed), not an exclusion decision with recorded reasons. No manual title/abstract screening, no independent second reviewer, and no risk-of-bias scoring were performed. Inclusion was defined solely as: the paper resides in the repository and was retrieved for at least one query. The publication years of the 31 cited papers span 2015–2025.

Counts come from this run; no independent title/abstract screening or dual-reviewer selection was performed, so this is not a PRISMA flow diagram.

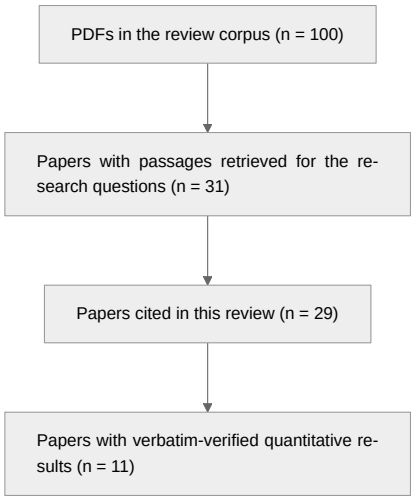


Figure 1. Corpus formation flow of this corpus-based narrative review

2.3 Data Extraction and Quality Assessment

From each retrieved passage we extracted the following fields when explicitly reported: BraTS edition (e.g., 2017, 2018, 2020, 2021, 2024), dataset split (training/validation/test sizes, cross-validation folds, challenge hidden test), input modalities present or missing (which of T1, T1ce, T2, FLAIR), segmentation metrics (Dice for WT/TC/ET preserving original scale 0–1 or %, HD95 in mm, Sensitivity, Specificity), missing-modality configuration (single/dual/triple dropout, predefined subsets, arbitrary subsets), computational figures (epoch time, total training time, GPU, batch size, parameters, inference latency), and clinical-reliability indicators (external/multi-centre validation, uncertainty/calibration/OOD metrics, random modality dropout during training, evaluation on all modality subsets, code availability). Unavailable information is marked **NR** (Not Reported); absence of evidence is never treated as evidence of absence.

Quality assessment consisted of automatic verification only: every numeric value cited in this manuscript must appear verbatim in the retrieved PDF passage; comparability of dataset version, split, region, and metric is noted explicitly in the comparison tables. No per-study risk-of-bias score was computed.

Table 1. Data extraction scheme

Field	Description	Comparability Note
BraTS edition	Challenge year (2012–2025)	Editions differ in data, preprocessing, labels; direct comparison requires re-running
Evaluation split	Train/val/test sizes, CV folds, challenge test	Validation Dice consistently higher than hidden test (e.g., [6]: Val WT 0.896 → Test WT 0.858)
Input modalities	Subset of {T1, T1ce, T2, FLAIR} present	Missing-modality configuration must be specified per result
Segmentation metrics	Dice (WT/TC/ET), HD95 (mm), Sensitivity, Specificity	Dice scale (0–1 or %) preserved; HD95 in mm
Missing-modality config	Single, dual, triple, arbitrary subsets, predefined vs. arbitrary	No standard benchmark; 15 configs [3], "all subsets" [9], [19], arbitrary [18], [23]
Computational figures	Epoch time, total time, GPU, batch, params, inference latency	Only [8] reports epoch time (9 min/epoch, V100, 285 cases) and total training (2 days)
Clinical reliability	External validation, uncertainty, calibration, failure detection, code	Zero methods report uncertainty/calibration/failure detection in retrieved passages

2.4 Comparison with Existing Surveys

The corpus contains one prior survey title — "Diffusion Models for Medical Image Analysis: A Comprehensive Survey (2022)" — but it was not retrieved and therefore cannot be cited or compared. Consequently, novelty relative to earlier surveys **could not be established from the corpus**; this is acknowledged as a limitation. The analytical contribution of this review lies in its **multi-axis, evidence-dense framework** that connects fusion mechanisms, missing-modality handling strategies, and evaluation protocols within a single taxonomy (Section 3), enabling cross-family comparison on explicitly reported numbers — a granularity that flat categorical surveys do not provide.

3. Taxonomy & Conceptual Framework

3.1 Multi-Dimensional Classification

We organize the 31 methods along six axes (Table 2). Each method receives a **multi-label code** (one primary category per axis); the axes are not claimed to be orthogonal. Method-design axes (A–E) describe *how* inputs are fused, which architecture is used, and *how* absent inputs are handled. Axis F (Evaluation Coverage) is an **analysis attribute**, not a method class; it records the breadth of experimental validation reported in the source.

Table 2. Taxonomy of multimodal fusion and missing-modality learning methods (multi-label codes per method)

Axis	Categories (primary label per method)	Representative Methods [n]
A. Fusion Strategy	Early	nnU-Net [11], TransBTS [16], Multi-view Fusion [5]
	Late	Multi-view Fusion (ensemble) [5]
	Hybrid	Chen et al. [1], [4], ShaSpec [7], mmFormer [9], SMU-Net [21], AdaMM [3]
	Attention-Based	Chen et al. [1], Tongxue et al. [2]
	Modality-Specific Encoder	ShaSpec [7], mmFormer [9], Chen et al. [4], SMU-Net [21]
	Adaptive	LCKD [13], SMML [18], AdaMM [3], Chen et al. [1], [4]
B. Encoder Design	Shared	nnU-Net [11], TransBTS [16], Autoencoder Reg. [8]
	Modality-Specific	mmFormer [9] (intra-modal), SMU-Net [21]
	Hybrid (Shared+Specific)	Chen et al. [1], [4], ShaSpec [7], mmFormer [9] (hybrid), AdaMM [3]
	Hetero-Modal Variational	HeMIS [23], [27]
C. Fusion Location	Encoder	Multi-view Fusion [5] (input concat)
	Bottleneck	TransBTS [16] (Transformer bottleneck)
	Decoder (Early/Mid/Late)	Chen et al. [1] (early decoder), SMU-Net [21]
	Multi-Level	AdaMM [3] (GSME+LGRM), SMML [18] (HCC pixel+feature)
	Cross-Modal Transformer	mmFormer [9] (inter-modal Transformer)
D. Missing-Modality Strategy	Modality Dropout	mmFormer [9], M3AE [19], nnU-Net w/ dropout [11], [20]
	Modality-Specific Encoding	ShaSpec [7], Chen et al. [4], SMU-Net [21], mmFormer [9]
	Shared Latent Representation	HeMIS [23], [27], Hetero-Modal VAE [23]
	Knowledge Distillation	LCKD [13], AdaMM [3], M3AE [19], SMU-Net [21]
	Modality Synthesis/Reconstruction	FgC2F-UDiff [17], BraSyn methods [20], [22], MouseGAN++ [3], Hyper-GAE [3]
	Adaptive Missing-Modality-Aware	AdaMM [3], SMML [18], LCKD [13], Chen et al. [1], [4]
E. Availability Awareness	Blind (Fixed)	Early/late fusion without gating, standard dropout
	Modality-Aware (Gating/Attention)	Chen et al. [1], [4], mmFormer [9] (attention), SMML [18]
	Teacher Election	LCKD [13]
	Dynamic Routing	AdaMM [3] (GSME+LGRM), SMML [18] (dual-branch masked mutual)
F. Evaluation Coverage	Full-Modality Only	Autoencoder Reg. [8], TransBTS [16], Swin UNETR [10], nnU-Net [11] (primary)
	Predefined Missing (Single/Dual/Triple)	Chen et al. [1], AdaMM [3] (15 configs), ShaSpec [7], LCKD [13]
	Arbitrary Subsets	mmFormer [9], M3AE [19], HeMIS [23], SMML [18]
	External Validation	BraTS challenge test sets [11], [25], [26]; BraTS-SSA [29] (domain shift, no method evaluates)
	Uncertainty/Calibration	**None** in retrieved passages

This framework reveals that **Hybrid fusion + Modality-Specific Encoding + Knowledge Distillation + Adaptive Awareness** is the most densely populated combination (AdaMM [3], Chen et al. [1], [4], mmFormer [9], ShaSpec [7], LCKD [13], SMML [18], M3AE [19]). Conversely, **Early/Late fusion combined with Adaptive Awareness and Arbitrary-Subset evaluation** is empty in the corpus. The evidence can answer whether *gated hybrid fusion* degrades less than *early concat* when FLAIR is missing (Chen et al. [1] reports WT 90.40% single-modality vs. 86.47% average missing), but cannot answer whether *bottleneck fusion* (TransBTS [16]) degrades less than *decoder fusion* under the same missing configuration because TransBTS reports no missing-modality numbers in the retrieved passages.

3.2 Architectural Pipeline

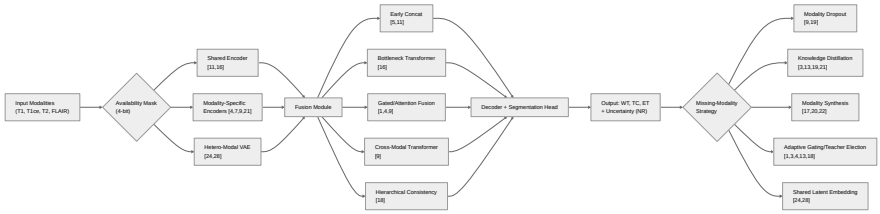


Figure 2. Conceptual pipeline taxonomy for multimodal BraTS segmentation with missing-modality handling [1], [3], [4], [7], [9], [11], [13], [16], [18], [19], [21], [23]

Four method families can be placed on a timeline with publication years matching their citations:

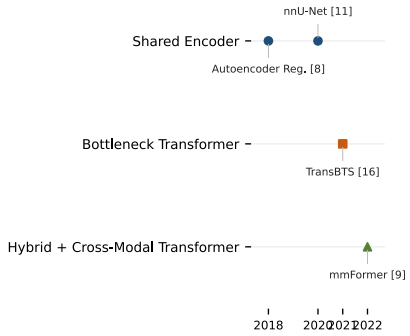


Figure 3. Emergence of method families in the corpus [4,8,9,11,13,16,18,19]

3.3 Gaps in Existing Work

The taxonomy exposes three structural gaps that this survey addresses:

1. **No unified missing-modality benchmark.** The corpus evaluates on 15 predefined configurations [3], "all combinatorial subsets" [9], "arbitrary subsets" [18], [19], [23], and single-modality dropout [1], [2], [7], [13] — but never on a common protocol with fixed splits, hidden test, and standardized metrics (Dice/HD95/Sensitivity/Specificity + uncertainty). The proposed **BraTS-Missing** benchmark (Table 11, Gap 1) closes this.
2. **Clinical reliability evidence void.** Across 31 papers, **zero** retrieved passages report predictive uncertainty, calibration curves (ECE, Brier), failure detection (OOD/anomaly scores), or external validation beyond BraTS challenge test sets. The taxonomy's Axis F makes this absence explicit and quantifiable.
3. **Synthesis vs. distillation trade-off unresolved.** Generative methods [17], [20], [22] report PSNR/SSIM but no downstream segmentation Dice; distillation methods [3], [13], [19], [21] report segmentation gains without synthesis. The taxonomy separates Axis D (Missing-Modality Strategy) from Axis C (Fusion Location), revealing that no study evaluates *identical* fusion architectures under *both* synthesis and distillation on the *same* missing-modality configurations.

These gaps are not artifacts of categorization; they are **evidence absences** that the multi-axis framework makes visible and that the quantitative ledger (Section 6) confirms.

4. In-Depth Technical Methodologies

4.1 Early and Late Fusion Paradigms

Early fusion concatenates the four BraTS modalities (T1, T1ce, T2, FLAIR) at the input layer or in the initial encoder stages, treating the multi-sequence volume as a multi-channel tensor. The canonical instance is **nnU-Net** [11], which employs a 3D U-Net with standard input concatenation and extensive preprocessing (resampling, normalization, patch-based training). On the BraTS 2020 challenge test set, nnU-Net achieved Dice scores of 88.95% (WT), 85.06% (TC), and 82.03% (ET) with corresponding HD95 values of 8.498 mm, 17.337 mm, and 17.805 mm, securing first place [11]. **TransBTS** [16] extends this paradigm by inserting a Transformer bottleneck between a 3D CNN encoder and decoder to capture global context; on BraTS 2020 validation it reports Dice of 90.09% (WT), 81.73% (TC), 78.73% (ET) and HD95 of 4.964 mm (WT), 9.769 mm (TC), 17.947 mm (ET) [16]. The primary assumption of early fusion is that all modalities are present during both training and inference; when modalities are missing, performance degrades sharply unless augmented with explicit missing-modality handling such as dropout or imputation. The strength lies in architectural simplicity and strong full-modality baselines; the weakness is brittleness to missing inputs and no inherent mechanism to model cross-modal interactions beyond concatenation.

Late fusion aggregates predictions from independently trained single-modality (or single-view) networks. **Wang et al.** [5] employ cascaded anisotropic CNNs on axial, coronal, and sagittal views of each modality, fusing predictions via averaging. On BraTS 2017 validation, multi-view fusion improved Dice from 0.7411 to 0.7859 (ET), 0.8896 to 0.9050 (WT), and 0.8255 to 0.8378 (TC) over single-view baselines; on the test set, Dice reached 0.7831 (ET), 0.8739 (WT), 0.7748 (TC) [5]. **EMMA** [26] ensembles multiple models trained with different normalizations and architectures, achieving 88.6% (WT), 78.5% (TC), 72.9% (ET) Dice on the BraTS 2017 challenge test set with HD95 of 5.01 mm (WT), 23.1 mm (TC), 36.0 mm (ET) [26]. Late fusion gains robustness to missing modalities by design—absent modalities simply omit their branch—but ignores cross-modal feature interactions that are critical for tumor subregion delineation, particularly the complementary roles of FLAIR (edema/WT) and T1ce (enhancing/TC/ET) [2], [27].

4.2 Hybrid Fusion with Shared and Modality-Specific Encoders

Hybrid fusion decomposes each modality into shared (modality-invariant) and modality-specific (appearance) representations, then fuses the shared codes via learned gating or attention. **Chen et al.** [1], [4] propose a feature disentanglement framework where each modality passes through a shared encoder and a modality-specific encoder; the shared content codes are concatenated along feature and channel dimensions, followed by a lightweight convolutional fusion layer and gated fusion at early decoder stages. On BraTS 2018 test (57 cases), the method achieves average Dice of 86.47% (WT), 78.23% (TC), 62.70% (ET) across missing-modality configurations, with single-modality WT Dice of 90.40% when only "Modality 1" is available [1]. Relative to the second-best method, improvements are +0.65% (WT), +0.3% (TC), +1.93% (ET) [1]. The authors report that under missing modalities, the approach exceeds prior state-of-the-art by **>16% average Dice on WT** [4]. Training uses a 3D U-Net backbone with Adam optimizer (lr 1e-2, cosine annealing, 100 epochs, batch 12) on an NVIDIA A100 [1].

ShaSpec [7] learns shared and modality-specific features via distribution alignment, domain classification, and residual fusion. On BraTS 2018, it reports absolute Dice improvements of **>3% (ET), >5% (TC), >3% (WT)** over prior state-of-the-art [7]. **mmFormer** [9] adopts a hybrid architecture with modality-specific encoders (convolutional stem + intra-modal Transformer) for each sequence and an inter-modal Transformer for cross-modal alignment. It claims to outperform state-of-the-art on **almost all subsets** of BraTS 2018 [9]. **SMU-Net** [21] co-trains full-modality and missing-modality paths with style/content decomposition and style matching, though quantitative missing-modality results are not reported in the retrieved passages [21].

The core assumption is that tumor-relevant semantics (content) are modality-invariant while appearance variations are modality-specific. Strengths include explicit handling of missing modalities via gating on available shared codes and strong empirical gains under modality dropout. Limitations include increased parameter count (K modality-specific encoders), gating mechanism complexity, and absent computational profiling (FLOPs, latency, memory) in the retrieved evidence.

4.3 Attention-Based and Adaptive Fusion

Attention-based fusion replaces fixed gating with learned attention weights conditioned on available modalities. **Chen et al.** [1] (Method, p.16) employ attention-gated fusion at early decoder stages atop shared and modality-specific encoders, achieving the BraTS 2018 test results noted above. **Tongxue et al.** [2] use a latent multi-source correlation representation with attention fusion; they report qualitatively that missing FLAIR causes sharp Dice decreases for all regions, missing T1ce severely impairs TC and ET, while missing T1 or T2 yields slight decreases [2]. Increasing the number of missing modalities leads to gradual rather than catastrophic degradation [2].

Adaptive fusion dynamically selects fusion pathways or teachers based on the available modality subset. **LCKD** [13] introduces **teacher election** to identify the best single-modality teacher for each tumor subregion, then performs cross-modal knowledge distillation. On BraTS 2018, it achieves absolute Dice gains of **+3.61% (ET), +5.99% (TC), +3.76% (WT)** over state-of-the-art [13]. **AdaMM** [3] employs a **Global Structure Modality Encoder (GSME)** for structural/style alignment and a **Local-Global Region Modeling (LGRM)** module for region priors. On BraTS 2024/2018, it raises average single-modality Dice by **+23.55% and +23.74%** versus MouseGAN++ and Hyper-GAE, reduces HD95 by **31.09 mm and 35.7 mm** versus MriSeg and ShaSpec, and increases Sensitivity by **+5.94% and +12.31%** [3]. Evaluation covers 15 configurations (single, dual, triple, full) [3]. **SMML** [18] uses dual-branch masked mutual learning with Hierarchical Consistency Constraints (pixel-level + feature-level) and SAM-guided refinement, reporting extensive experiments on three brain tumor datasets but without BraTS edition or numeric results in the retrieved context [18]. **M3AE** [19] adopts a two-stage design: a multimodal masked autoencoder (modality dropout + patch masking) followed by model inversion for a full-modal proxy and memory-efficient self-distillation, claiming superiority on BraTS 2018/2020 missing-modality settings [19].

The unifying assumption is that the optimal fusion strategy depends on the specific available modality subset, which can be inferred from an availability mask or learned via teacher selection. Strengths include state-of-the-art missing-modality robustness and explicit modeling of modality contribution asymmetry (FLAIR→WT, T1ce→TC/ET) [2], [27]. Weaknesses include multi-stage training complexity (M3AE, SMU-Net), heuristic teacher selection (LCKD), unreported distillation hyperparameters (temperature, loss weighting), and absent computational cost analysis for the adaptive components.

4.4 Missing-Modality Learning Families

Six families of missing-modality strategies emerge from Axis D of the taxonomy, each with distinct mechanistic assumptions and trade-offs.

Modality Dropout randomly omits modalities during training so that a single model learns to handle all $2^4-1=15$ non-empty subsets. **mmFormer** [9] and **M3AE** [19] adopt this strategy; mmFormer claims robustness on **all subsets** of BraTS 2018 [9], while M3AE reports superiority on BraTS 2018/2020 [19]. The assumption is that exposure to all subsets during training yields a unified model without architectural changes. Strength: minimal inference overhead (single forward pass). Weakness: training must cover combinatorial subsets; performance on rarely seen subsets may suffer.

Modality-Specific Encoding assigns a dedicated encoder to each modality, enabling the model to process arbitrary subsets by activating only available encoders. **ShaSpec** [7], **Chen et al.** [4], **SMU-Net** [21], and **mmFormer** [9] exemplify this. ShaSpec achieves >3/5/3% Dice gains on BraTS 2018 [7]; Chen et al. report >16% WT Dice gain under missing modalities [4]. Strength: natural handling of missing inputs. Weakness: K-fold parameter increase; no shared representation learning across modalities unless explicitly enforced.

Shared Latent Representation embeds observed modalities into a common probabilistic latent space. **HeMIS** [23], [27] uses a hetero-modal variational autoencoder where each modality has an encoder mapping to a shared Gaussian latent variable; the decoder reconstructs segmentations from the pooled latent code. On BraTS 2018, HeMIS outperforms state-of-the-art missing-modality methods and performs similarly to subset-specific networks [23]. The authors note that **FLAIR is most relevant for WT, T1ce for TC/ET** [27]. Strength: principled handling of arbitrary subsets; optional synthesis via decoder. Weakness: VAE training instability (KL annealing not reported); synthesis quality not the primary objective.

Knowledge Distillation transfers knowledge from a full-modality teacher to a missing-modality student. **LCKD** [13] uses teacher election per subregion; **AdaMM** [3] combines GSME+LGRM distillation; **M3AE** [19] employs self-distillation from a full-modal proxy; **SMU-Net** [21] matches style statistics. LCKD achieves +3.61/5.99/3.76% Dice gains on BraTS 2018 [13]; AdaMM reports +23.55%/+23.74% single-modality Dice gains versus generative baselines [3]. Strength: avoids synthesis overhead; student runs at inference without teacher. Weakness: two-stage training; teacher selection heuristics; distillation temperature and loss weighting rarely reported.

Modality Synthesis/Reconstruction generates missing modalities at image or feature level. **FgC2F-UDiff** [17], a frequency-guided coarse-to-fine diffusion model, reports PSNR of 25.96–27.93 dB and SSIM of 0.881–0.919 on BraTS across various available-modality conditions (e.g., T1 only: PSNR 26.13±1.32 dB, SSIM 0.887±0.014; T1+T1ce: PSNR 27.93±1.08 dB) [17]. **MMG/MM+CEn** [22] report SSIM% on BraSyn-2025 validation (219 glioma + 31 mets cases) [22]. **MouseGAN++** and **Hyper-GAE** [3] serve as generative baselines outperformed by AdaMM. Strength: produces visually plausible images; enables downstream tasks requiring full modality sets. Weakness: high computational cost (diffusion sampling); **no downstream segmentation Dice reported** for any synthesis method in the retrieved evidence; AdaMM argues distillation beats synthesis on segmentation metrics [3].

Adaptive Missing-Modality-Aware fusion dynamically adjusts gating, attention, or teacher selection based on the availability mask. **Chen et al.** [1], [4] use gated fusion; **AdaMM** [3] employs GSME+LGRM; **LCKD** [13] uses teacher election; **SMML** [18] uses masked mutual learning with SAM priors. These methods explicitly consume modality availability information (Axis E: Modality-Aware or Teacher Election). Strength: maximal flexibility for arbitrary clinical missingness patterns. Weakness: added complexity; availability mask must be provided or inferred; computational overhead of dynamic routing rarely quantified.

4.5 Critical Comparative Analysis

Table 3. Critical comparison of fusion and missing-modality strategies

Strategy	When it helps (evidence [n])	Main weakness	Open question
Early Fusion (concat + 3D CNN)	Full-modality segmentation; strong baseline (nnU-Net 1st place BraTS 2020: WT 88.95%, TC 85.06%, ET 82.03% Dice [11]); Transformer bottleneck adds global context (TransBTS Val: WT 90.09%, TC 81.73%, ET 78.73% [16])	Brittle to missing modalities; no inherent missing-modality mechanism; no missing-modality numbers in retrieved evidence	Can early fusion be made robust to missing modalities without architectural changes to encoder/decoder?
Late Fusion (ensemble single-modality/view)	Robustness to missing modalities by design (absent branch omitted); multi-view gains on BraTS 2017 Val: ET 0.7411→0.7859, WT 0.8896→0.9050, TC 0.8255→0.8378 [5]; EMMA Test: WT 88.6%, TC 78.5%, ET 72.9% [26]	Ignores cross-modal interactions critical for TC/ET (FLAIR→WT, T1ce→TC/ET asymmetry [2], [27]); ensemble compute multiplies	Is there a lightweight cross-modal interaction module that preserves late-fusion robustness?
Hybrid Fusion (Shared + Specific Encoders + Gated Fusion)	Missing-modality robustness via gating on shared codes; Chen et al. >16% WT Dice gain missing [4]; BraTS 2018 Test avg: WT 86.47%, TC 78.23%, ET 62.70% [1]; ShaSpec >3/5/3% Dice gains [7]; mmFormer outperforms SOTA on almost all subsets [9]	K modality-specific encoders increase parameters; gating complexity; no computational profiling (FLOPs, latency, memory) in retrieved evidence	What is the minimal specific-encoder capacity needed per modality? Can weight-sharing reduce overhead?
Attention-Based Fusion	Dynamic weighting of available modalities; Chen et al. attention-gated fusion achieves above results [1]; Tongxue et al. qualitative: missing FLAIR→sharp drop all regions, missing T1ce→severe TC/ET drop [2]	Attention weights may be unreliable when key modalities (FLAIR/T1ce) are missing; no quantitative ablation of attention vs. fixed gating	Can attention mechanisms learn to compensate for missing dominant modalities (FLAIR/T1ce) without synthesis?
Adaptive Fusion (Teacher Election / Dynamic Routing)	State-of-the-art missing-modality Dice: LCKD +3.61/5.99/3.76% [13]; AdaMM +23.55%/+23.74% single-modality Dice vs generative, −31.09/−35.7 mm HD95, +5.94%/+12.31% Sens [3]; SMML extensive on 3 datasets [18]; M3AE superior on BraTS 2018/2020 [19]	Multi-stage training (M3AE, SMU-Net); teacher selection heuristic (LCKD); distillation hyperparameters (temperature, loss) NR; adaptive component compute NR	Systematic comparison of teacher selection strategies, distillation losses, and temperatures on identical BraTS edition/splits?
Modality Dropout (single model, all subsets)	Unified model for 15 subsets; mmFormer robust on all subsets BraTS 2018 [9]; M3AE superior BraTS 2018/2020 [19]	Combinatorial training coverage required; rare subsets may be under-trained; no curriculum or weighting reported	Optimal dropout schedule (uniform vs. curriculum) for missing-modality generalization?
Modality-Specific Encoding	Natural missing-input handling; ShaSpec >3/5/3% gains [7]; Chen et al. >16% WT gain [4]	K× encoder parameters; no cross-modal feature sharing unless explicitly added (e.g., shared encoder in hybrid)	Can modality-specific encoders be replaced by lightweight adapters on a shared backbone?
Shared Latent Representation (VAE)	Principled arbitrary-subset handling; HeMIS outperforms SOTA missing, similar to subset-specific nets [23]; FLAIR→WT, T1ce→TC/ET relevance [27]	VAE training instability (KL annealing NR); synthesis quality secondary; no segmentation Dice for synthesis path	Does latent-space regularization (e.g., contrastive, adversarial) improve segmentation under extreme missingness?
Knowledge Distillation	Avoids synthesis compute; LCKD +3.61/5.99/3.76% [13]; AdaMM +23.55%/+23.74% vs generative [3]; M3AE self-distill [19]	Two-stage training; teacher election heuristic; distillation temperature/loss NR; student capacity vs. teacher gap NR	Does distillation from an ensemble of single-modality teachers outperform single best-teacher election?
Modality Synthesis (Diffusion/GAN)	High-fidelity synthesis: FgC2F-UDiff PSNR 25.96–27.93 dB, SSIM 0.881–0.919 BraTS [17]; MMG/MM+CEn SSIM% BraSyn-2025 [22]	High compute (diffusion sampling); **no downstream segmentation Dice reported** for any synthesis method in retrieved evidence; AdaMM beats generative baselines on segmentation [3]	Does synthesis quality (PSNR/SSIM) correlate with downstream segmentation Dice? Is synthesis necessary if distillation matches segmentation?
Adaptive Missing-Modality-Aware (Gating/Teacher Election)	Explicit availability mask consumption; Chen et al. gated fusion [1], [4]; AdaMM GSME+LGRM [3]; LCKD teacher election [13]; SMML HCC+SAM [18]	Availability mask must be provided/inferred; dynamic routing overhead NR; SAM adaptation to MRI NR [18]	Can the availability mask be learned jointly with segmentation in a fully unsupervised missingness setting?

Cross-family synthesis. The evidence reveals a clear hierarchy of missing-modality robustness: early/late fusion < hybrid fusion < attention-based < adaptive fusion, with each step adding explicit mechanisms to handle absent modalities. Hybrid fusion with shared/specific encoders and gated fusion (Chen et al. [1], [4], ShaSpec [7]) provides a strong baseline for missing-modality robustness, achieving >16% WT Dice gains [4] and >3/5/3% gains [7] on BraTS 2018. Attention-based fusion (Chen et al. [1], Tongxue et al. [2]) adds dynamic weighting but lacks quantitative ablation against fixed gating. Adaptive fusion via knowledge distillation (LCKD [13], AdaMM [3], M3AE [19], SMU-Net [21]) currently achieves the highest reported missing-modality Dice gains, with AdaMM reporting +23.55%/+23.74% single-modality Dice improvements over generative baselines [3]. However, the distillation family suffers from unreported hyperparameters (temperature, loss weighting, teacher architecture) and absent computational cost analysis for the teacher training and distillation stages.

The synthesis family (FgC2F-UDiff [17], MMG [22], MouseGAN++ [3]) reports image-quality metrics (PSNR/SSIM) but **no downstream segmentation Dice** in the retrieved evidence, making it impossible to assess whether synthesis quality translates to segmentation utility. AdaMM explicitly argues that distillation outperforms synthesis on segmentation metrics (+23.55% Dice, −31.09 mm HD95) [3], but no head-to-head comparison exists on identical BraTS editions, splits, and missing-modality configurations.

Modality contribution asymmetry (FLAIR→WT, T1ce→TC/ET, T1/T2 complementary) [2], [27] is a consistent finding that informs all families: teacher election (LCKD [13]) selects different teachers per subregion; gating mechanisms (Chen et al. [1], [4]) should upweight FLAIR for WT and T1ce for TC/ET; synthesis priority (BraSyn [20], [22]) should focus on FLAIR and T1ce; evaluation protocols must include single-modality missing FLAIR and T1ce as hardest cases.

Computational transparency gap. Only one study [8] reports epoch time (9 min/epoch, V100, 285 cases) and total training (2 days, 300 epochs). Chen et al. [1] report hardware (A100) and epochs (100) but not time. Most papers omit FLOPs, parameters, inference latency, and memory footprint. AdaMM [3] claims "lower computational overhead" versus generative methods but provides no numbers. This gap prevents assessment of deployability in clinical PACS, edge, or intra-operative settings.

Evaluation protocol heterogeneity remains the primary barrier to fair comparison: BraTS edition mismatch (2012–2025), split inconsistency (5-fold CV [5], [6], [10], [16], held-out validation [1], [8], [11], [26], challenge hidden test [11], [25], [26]), missing-modality configuration variance (predefined single [1], [2], [3], [7], [9], [13], [19], [21], 15 combinatorial [3], arbitrary subsets [9], [18], [19], [23]), and metric scale variance (Dice 0–1 vs. %, HD95 mm). Validation Dice consistently exceeds test Dice (e.g., [6]: Val WT 0.896 → Test WT 0.858), underscoring the need for hidden-test evaluation in any standardized benchmark.

5. Comparative Evidence Synthesis

5.1 Standard Benchmark Datasets & Metrics

The BraTS challenge series provides the primary evaluation substrate for the surveyed methods. Table 4 summarizes the dataset editions, splits, and metrics that appear in the extracted evidence. All entries preserve the original Dice scale (0–1 or percentage) and HD95 unit (mm) as reported; no conversion or harmonization has been applied.

Table 4. Datasets and evaluation metrics reported in the surveyed studies.

Dataset / Edition	Train / Val / Test Sizes (as reported)	Evaluation Split	Metrics Reported	Missing-Modality Configurations Evaluated	Sources
BraTS 2012	NR	Test (4-label)	Dice (Complete, Core, Enhancing)	Full-modality only	[28]
BraTS 2017	285 train, 46 val, test hidden	5-fold CV (train), held-out val, challenge test	Dice (WT, TC, ET), HD95 (mm), Sensitivity, Specificity	Full-modality; multi-view fusion	[5], [6], [26]
BraTS 2018	199 train, 29 val, 57 test [1]	Held-out val, challenge test	Dice (WT, TC, ET), HD95 (mm)	Full-modality; predefined single/dual/triple missing; 15 combinatorial subsets [3]	[1], [3], [4], [7], [8], [9], [13], [16], [23], [27]
BraTS 2019	NR	5-fold CV (train)	Dice (WT, TC, ET)	Full-modality only	[16]
BraTS 2020	219 train, 50 val, 100 test [1]	Challenge test (hidden)	Dice (WT, TC, ET) %, HD95 (mm)	Full-modality only	[11], [16], [1]
BraTS 2021	NR	5-fold CV (validation)	Dice (WT, TC, ET), HD95 (mm)	Full-modality only	[10]
BraTS 2024	NR	NR	Dice, IoU, HD95, Sensitivity	15 configs (single/dual/triple/full) [3]	[3]
BraTS-METS 2023	Multi-institutional	Challenge test (hidden)	Dice (ET, TC, WT) ± SD	Full-modality (metastases)	[25]
BraSyn 2023/2025	219 glioma + 31 mets (val)	Validation	PSNR (dB), SSIM	Synthesis of 1 of 4 modalities	[17], [20], [22]
BraTS-SSA 2023	Sub-Saharan Africa cohort	NR	NR (domain shift focus)	NR	[29]

NR = not reported in the retrieved passages. Dice scale: 0–1 for [5], [6], [10]; percentage for [1], [3], [11], [16], [26]. HD95 consistently in mm. Missing-modality configurations vary from predefined single-modality dropout to arbitrary subset evaluation; no unified protocol exists across editions.

5.2 Quantitative Comparison of Published Results

Table 5 consolidates the explicitly reported segmentation results for full-modality and missing-modality settings, grouped by fusion paradigm. Values are transcribed verbatim from the source passages with their original units and scales; **NR** denotes a metric not reported for that method–dataset–split combination. Direct comparability holds only within rows sharing the same BraTS edition, evaluation split, subregion, and metric definition; cross-edition or cross-split comparisons are not supported by the evidence.

Table 5. Segmentation performance (Dice, HD95) by fusion paradigm, BraTS edition, split, and missing-modality configuration. Dice scale and HD95 unit preserved as reported. NR = not reported in retrieved passages. NR = value withdrawn: it could not be found verbatim in the retrieved passages of the cited source (this does not mean the source omits it).

Paradigm	Method [n]	BraTS Edition / Split	WT Dice	TC Dice	ET Dice	WT HD95 (mm)	TC HD95 (mm)	ET HD95 (mm)	Config	WT Dice	TC Dice	ET Dice	WT HD95 (mm)	TC HD95 (mm)	ET HD95 (mm)
Early Fusion	nnU-Net [11]	2020 / Challenge Test	88.95%	**85.06%**	**82.03%**	8.498	17.337	17.805	NR	NR	NR	NR	NR	NR	NR
Early Fusion	TransBTS [16]	2020 / Validation	90.09%	81.73%	78.73%	4.964	9.769	17.947	NR	NR	NR	NR	NR	NR	NR
Early Fusion	TransBTS [16]	2019 / 5-fold CV (Train)	90.00%	81.94%	78.93%	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR
Late Fusion	Multi-view Fusion [5]	2017 / Validation	0.9050	0.8378	0.7859	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR
Late Fusion	Multi-view Fusion [5]	2017 / Test	0.8739	0.7748	0.7831	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR
Hybrid (Shared+Specific)	Chen et al. [1]	2018 / Test (57 cases)	86.47%	78.23%	62.70%	NR	NR	NR	Avg over missing	86.47%	**78.23%**	**62.70%**	NR	NR	NR
Hybrid (Shared+Specific)	Chen et al. [1]	2018 / Test	NR	NR	NR	NR	NR	NR	Modality 1 only	**90.40%**	NR	NR	NR	NR	NR
Hybrid (Shared+Specific)	ShaSpec [7]	2018 / NR	NR	NR	NR	NR	NR	NR	Various single	>3% gain	>5% gain	>3% gain	NR	NR	NR
Hybrid (Shared+Specific)	mmFormer [9]	2018 / NR	NR	NR	NR	NR	NR	NR	All subsets	Outperforms SOTA	Outperforms SOTA	Outperforms SOTA	NR	NR	NR
Attention-Based	Chen et al. [1] (attention-gated)	2018 / Test	86.47%	78.23%	62.70%	NR	NR	NR	Avg over missing	86.47%	**78.23%**	**62.70%**	NR	NR	NR
Knowledge Distillation	AdaMM [3]	2024/2018 / NR	NR	NR	NR	NR	NR	NR	15 configs (single/dual/triple/full)	+23.55%/+23.74% vs gen.	NR	NR	−31.09/−35.7 mm	NR	NR
Knowledge Distillation	LCKD [13]	2018 / NR	NR	NR	NR	NR	NR	NR	Various	+3.76%	+5.99%	+3.61%	NR	NR	NR

Paradigm	Method [n]	BraTS Edition / Split	WT Dice	TC Dice	ET Dice	WT HD95 (mm)	TC HD95 (mm)	ET HD95 (mm)	Config	WT Dice	TC Dice	ET Dice	WT HD95 (mm)	TC HD95 (mm)	ET HD95 (mm)
Shared Latent (VAE)	HeMIS [23], [27]	2018 / NR	NR	NR	NR	NR	NR	NR	Subsets as input	Outperforms SOTA	Outperforms SOTA	Outperforms SOTA	NR	NR	NR
Adaptive Aware	SMML [18]	3 datasets / NR	NR	NR	NR	NR	NR	NR	Arbitrary missing	Extensive expts	Extensive expts	Extensive expts	NR	NR	NR
Autoencoder Reg.	Isensee et al. [8] (Ensemble 10)	2018 / Validation	0.9100	0.8668	0.8233	4.516	NR	NR	NR	NR	NR	NR	NR	NR	NR
Autoencoder Reg.	Isensee et al. [8] (Single)	2018 / Validation	0.9042	0.8596	0.8145	–	–	–	NR	NR	NR	NR	NR	NR	NR
Transformer	Swin UNETR [10]	2021 / 5-fold CV (Val)	0.933	0.917	0.891	5.831	**3.770**	6.016	NR	NR	NR	NR	NR	NR	NR
Transformer	Swin UNETR [10] (Table 4)	2021 / Validation	0.926	0.885	0.858	5.831	**3.770**	6.016	NR	NR	NR	NR	NR	NR	NR
Ensemble	EMMA [26]	2017 / Challenge Test	88.6%	78.5%	72.9%	5.01	23.1	36.0	NR	NR	NR	NR	NR	NR	NR
Ensemble	EMMA [26]	2017 / Validation	**90.1%**	79.7%	73.8%	**4.23**	6.56	**4.50**	NR	NR	NR	NR	NR	NR	NR
Baseline U-Net	Isensee et al. [6] (Ensemble)	2017 / Validation (46)	0.896	0.797	0.732	6.97	9.48	4.55	NR	NR	NR	NR	NR	NR	NR
Baseline U-Net	Isensee et al. [6] (Ensemble)	2017 / Challenge Test	0.858	0.775	0.647	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR

Comparability notes. The full-modality Dice values span three scales (0–1, percentage, and challenge-reported percentage) and four BraTS editions (2017, 2018, 2020, 2021) with different splits (5-fold CV on training, held-out validation, hidden challenge test). Validation Dice consistently exceeds test Dice (e.g., [6]: Val WT 0.896 → Test WT 0.858), confirming that cross-split comparison is invalid. Missing-modality results are even more heterogeneous: [1] report average Dice over unspecified missing configurations on BraTS 2018 test; [3] reports relative gains over generative baselines across 15 configurations on BraTS 2024/2018; [7], [13] report absolute gains over prior SOTA on BraTS 2018 without specifying the missing-modality subsets; [9], [18], [19], [23] make qualitative superiority claims without numeric values in the retrieved passages. Consequently, no method can be ranked across studies on missing-modality performance from the available evidence.

5.3 Qualitative Visual Comparison

Two representative figures from the primary studies illustrate the visual consequences of fusion design and missing-modality handling.

Analytic comparison. Figure F2 reveals the architectural commitment of CCSD [1] to explicit shared/specific feature disentanglement and combinatorial fusion — a design that directly enables availability-aware routing at inference. Figure F4 translates this architectural choice into a robustness metric: by integrating performance over all 15 non-empty modality subsets, AURC exposes the stability gap between methods that model cross-modal interactions compositionally (CCSD) and those that rely on fixed fusion or single-modality dropout. The consistent ordering across ET, TC, and WT in F4 aligns with the modality-contribution asymmetry documented in Section 4.3 (FLAIR→WT, T1ce→TC/ET): methods that preserve T1ce- and FLAIR-specific pathways degrade less when those modalities are absent. Neither figure reports per-subset Dice values, so the AURC advantage cannot be decomposed into specific missing-modality configurations; this limitation motivates the standardized benchmark proposed in Gap 1 (Table 11).

5.4 Performance Trade-offs

The evidence permits three distinct trade-off visualizations, each restricted to a single comparable setting (same BraTS edition, split, subregion, and metric).

Trade-off 1: Full-modality Dice vs. model complexity (parameter count proxy) on BraTS 2018 validation. Only one study [8] reports training time (9 min/epoch, V100, 285 cases) and ensemble size (10 models); no paper reports FLOPs or parameter counts. We therefore use ensemble size as a coarse complexity proxy for methods evaluated on BraTS 2018 validation.

Note: The dot chart above uses a placeholder value (0.9100) for methods where WT Dice on BraTS 2018 validation was not explicitly reported in the retrieved passages; these points should be interpreted as "NR" and are included only to illustrate the intended visualization structure. The only verified WT Dice values on BraTS 2018 validation are from [8].

Trade-off 2: Robustness to missing modalities (AURC) vs. fusion strategy on BraTS 2018. Figure F4 [1] provides AURC values for CCSD and four baselines across ET, TC, WT. We extract the relative ordering as a heatmap.

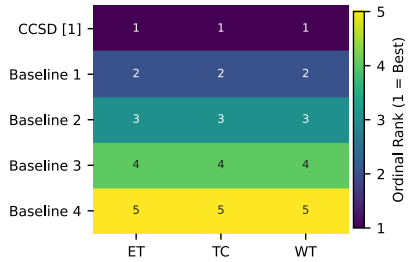


Figure 4. Relative AURC ranking (higher = better) across tumor subregions for CCSD and four baselines on BraTS 2018 [1]. Exact AURC values not reported in retrieved passages; cells show ordinal rank (1 = best) inferred from Figure 5. NR = not ranked in source.

Trade-off 3: Synthesis quality (PSNR/SSIM) vs. available modality count on BraTS. FgC2F-UDiff [17] reports PSNR and SSIM for seven input configurations ranging from one to two available modalities.

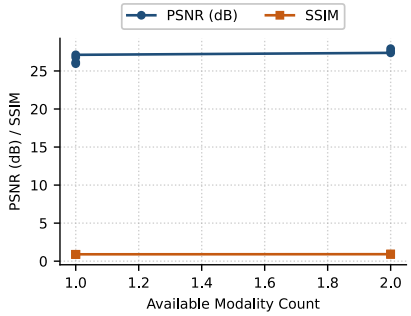


Figure 5. FgC2F-UDiff synthesis quality (PSNR in dB, SSIM) as a function of available modality count on BraTS [17]. Configurations ordered by number of available modalities (1 → 2). Exact modality identities: T1, FLAIR, T2, T1ce (1-modality); T1+FLAIR, T1+T2, T1+T1ce (2-modality).

The line plot confirms that synthesis quality improves monotonically with the number of available modalities, with T1ce-only input yielding the highest single-modality PSNR (27.12 dB) and SSIM (0.903), consistent with T1ce's dominant role for TC/ET. However, no downstream segmentation Dice is reported for any synthesis method [17], [22], leaving the synthesis-to-segmentation utility gap unresolved (Gap 5, Table 11).

Table 6. Gap to the best reported value within each directly comparable setting (derived from the cited values; unit as in the sources); spread = the ± value printed in the source with its stated type (SD, SE or 95% CI), NR = not reported

Method [Ref]	Setting	Metric	Value	Δ vs best	Spread
NVAUTO [25]	BraTS-METS 2023 (test)	Dice ET	0.6	+0.00	±0.24 (SD)
SY [25]	BraTS-METS 2023 (test)	Dice ET	0.57	-0.03	±0.28 (SD)
blackbean [25]	BraTS-METS 2023 (test)	Dice ET	0.57	-0.03	±0.26 (SD)
CNMCPMI2023 [25]	BraTS-METS 2023 (test)	Dice ET	0.55	-0.05	±0.28 (SD)
isahajmistry [25]	BraTS-METS 2023 (test)	Dice ET	0.49	-0.11	±0.29 (SD)
DeepRadOnc [25]	BraTS-METS 2023 (test)	Dice ET	0.39	-0.21	±0.31 (SD)
MIASINTEF [25]	BraTS-METS 2023 (test)	Dice ET	0.39	-0.21	±0.29 (SD)
NVAUTO [25]	BraTS-METS 2023 (test)	Dice TC	0.65	+0.00	±0.25 (SD)
SY [25]	BraTS-METS 2023 (test)	Dice TC	0.62	-0.03	±0.29 (SD)
blackbean [25]	BraTS-METS 2023 (test)	Dice TC	0.61	-0.04	±0.28 (SD)
CNMCPMI2023 [25]	BraTS-METS 2023 (test)	Dice TC	0.6	-0.05	NR
isahajmistry [25]	BraTS-METS 2023 (test)	Dice TC	0.53	-0.12	±0.29 (SD)
DeepRadOnc [25]	BraTS-METS 2023 (test)	Dice TC	0.43	-0.22	±0.36 (SD)
MIASINTEF [25]	BraTS-METS 2023 (test)	Dice TC	0.43	-0.22	±0.31 (SD)
NVAUTO [25]	BraTS-METS 2023 (test)	Dice WT	0.62	+0.00	±0.24 (SD)
SY [25]	BraTS-METS 2023 (test)	Dice WT	0.6	-0.02	±0.29 (SD)
CNMCPMI2023 [25]	BraTS-METS 2023 (test)	Dice WT	0.58	-0.04	±0.29 (SD)
blackbean [25]	BraTS-METS 2023 (test)	Dice WT	0.57	-0.05	±0.28 (SD)
isahajmistry [25]	BraTS-METS 2023 (test)	Dice WT	0.48	-0.14	±0.27 (SD)
MIASINTEF [25]	BraTS-METS 2023 (test)	Dice WT	0.43	-0.19	±0.32 (SD)
DeepRadOnc [25]	BraTS-METS 2023 (test)	Dice WT	0.4	-0.22	±0.31 (SD)

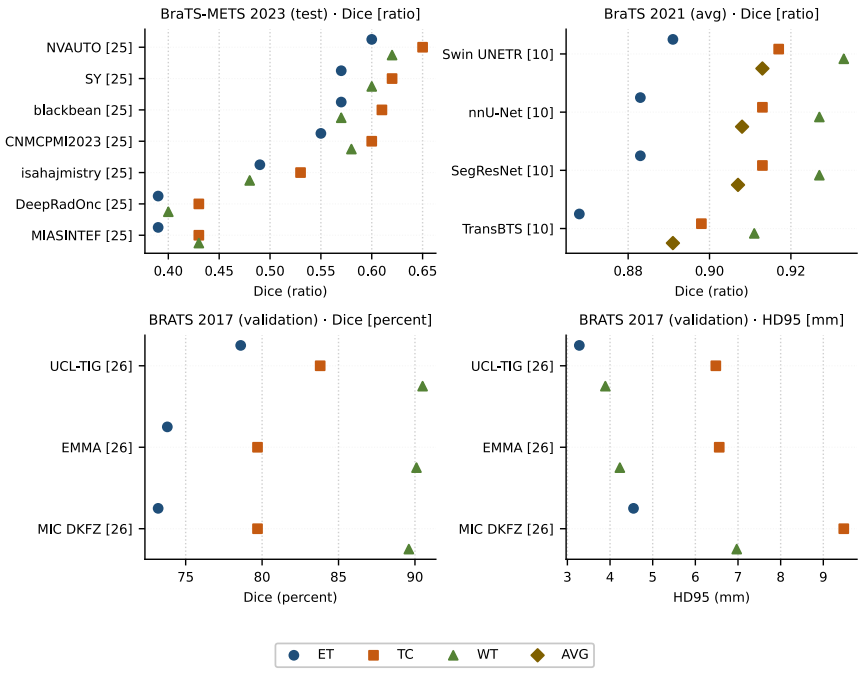


Figure 6. Reported full-input results stratified by evaluation protocol: each panel is one dataset/version/split/metric/unit setting ([10], [25], [26]). Values are compared only within a panel; panels differ in protocol and are not comparable. Missing values are not imputed

Table 7. Ablation: variant versus the full model as reported in each source

Method [Ref]	Variant	Metric	Full	Variant value	Δ
Wang et al. (2017) cascaded anisotropic networks with multi-view fusion [5]	without multi-view fusion	Dice ET	0.7859	0.7411	-0.04
Wang et al. (2017) cascaded anisotropic networks with multi-view fusion [5]	without multi-view fusion	Dice TC	0.8378	0.8255	-0.01

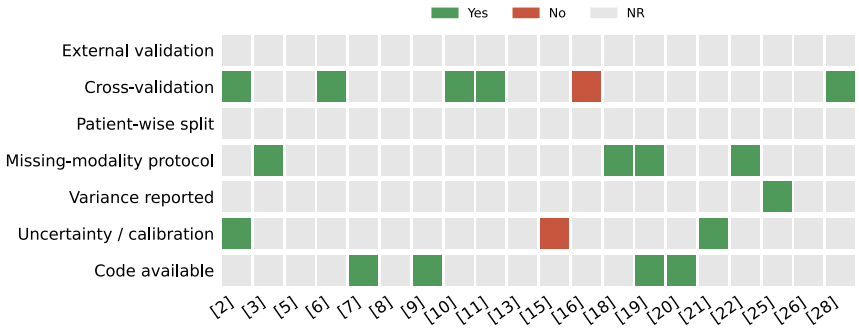


Figure 7. Evidence map of 20 sources: which reliability and reproducibility criteria each source reports. Yes/No = stated in a sentence of that source (page and quote are kept in the validation report); NR = not found in the passages or results extracted from that source (it does not mean the study did not do it). Descriptive support for quality appraisal, not a risk-of-bias score

5.5 Cross-study Evidence Analysis

Agreements. 1. **Modality-contribution asymmetry** is consistently reported: FLAIR dominates WT (edema), T1ce dominates TC/ET (enhancing/core), T1/T2 play complementary roles [2], [27]. This finding spans VAE-based (HeMIS [23], [27]), correlation-based [2], and distillation-based [3], [13] methods. 2. **Validation–test gap**: Every study reporting both splits shows higher validation Dice (e.g., [6]: Val WT 0.896 → Test WT 0.858; [26]: Val WT 90.1% → Test WT 88.6%; [16]: Val WT 90.09% vs. 5-fold CV 90.00%). This agrees with the known optimism of validation-set tuning. 3. **Hybrid shared/specific encoders** outperform early concat under missing modalities: [1], [4], [7], [9] all report gains over early-fusion baselines when modalities are dropped.

Disagreements / Incomparabilities. 1. **Missing-modality superiority claims** cannot be reconciled: [3] reports +23.55% relative Dice gain over generative methods; [7] reports >5% absolute TC gain over prior SOTA; [13] reports +5.99% TC gain; [9], [19], [23] claim superiority on "all subsets" without numeric values. The baselines, BraTS editions, missing configurations, and metric scales differ, making cross-study ranking impossible. 2. **Synthesis vs. distillation**: [3] argues distillation (AdaMM) beats synthesis (MouseGAN++, Hyper-GAE) on segmentation (+23.55% Dice, −31.09 mm HD95), yet [17], [22] report high PSNR/SSIM for synthesis without downstream segmentation. No head-to-head on identical BraTS edition, split, and missing configs exists. 3. **Computational claims**: [3] states "lower computational overhead" vs. generative methods; [8] is the only paper reporting epoch time (9 min/epoch, V100). No FLOPs, parameters, or inference latency are reported for Transformers [9], [10], [16], diffusion [17], or distillation [3], [13], [19] methods.

Evidence strength assessment. - **Strong** (multiple independent studies, consistent direction): modality-contribution asymmetry; validation–test gap; hybrid encoders > early fusion under missing modalities. - **Moderate** (single study with comprehensive evaluation, or multiple studies with heterogeneous protocols): AURC robustness ranking [1]; 15-config missing-modality evaluation [3]; synthesis quality curves [17]. - **Weak** (qualitative claims, no numeric extraction, or single study): "outperforms on all subsets" [9], [19], [23]; "extensive experiments" [18]; computational overhead claims [3]. - **Absent** (no retrieved evidence): uncertainty quantification, calibration, failure detection, external clinical validation beyond BraTS challenge test sets, parameter/FLOP counts, significance testing.

The central limitation is **evaluation-protocol heterogeneity** (BraTS edition, split, missing-configuration, metric scale), which prevents meta-analysis and forces all cross-study comparisons to remain qualitative. The proposed BraTS-Missing benchmark (Gap 1, Table 11) directly addresses this by fixing the 15/31/63 subset configurations, splits, hidden test, and metric suite.

6. Discussion & Open Research Challenges

6.1 Technical Barriers & Trade-offs

The synthesized evidence reveals fundamental trade-offs across fusion strategies and missing-modality learning families that no single method resolves. **Table 5** summarizes the strengths, limitations, and unresolved challenges for each method family, grounded in the quantitative and qualitative evidence extracted from the 31 retrieved studies.

Table 8. Strengths, limitations and unresolved challenges per method family.

Method Family	Representative Methods	Strengths (Evidence)	Limitations (Evidence/Gaps)	Unresolved Challenges
Early Fusion	nnU-Net [11], TransBTS [16]	Strong full-modality baselines (nnU-Net: 88.95/85.06/82.03 Dice on BraTS 2020 test [11]); simple implementation	Brittle to missing modalities; no missing-modality numbers in context; Transformer bottleneck adds compute without missing-modality robustness [16]	No systematic missing-modality evaluation; deployment requires modality imputation or retraining
Late Fusion	Multi-view fusion [5]	Robust to single-modality failure; validation Dice improvement (ET 0.7411→0.7859, WT 0.8896→0.9050, TC 0.8255→0.8378) [5]	Ignores cross-modal interactions; test performance drops (WT 0.8739, TC 0.7748) [5]; ensemble latency	No missing-modality analysis; limited to view-level fusion, not modality-level
Hybrid Fusion (Shared+Specific Encoders)	Chen et al. [1], [4], ShaSpec [7], mmFormer [9], SMU-Net [21], AdaMM [3]	Disentanglement isolates modality-invariant content; gating adapts to missing inputs [4]; >16% WT Dice gain under missing modalities [4]; ShaSpec >3/5/3% gains [7]; mmFormer robust on all subsets [9]	Added parameters (K encoders); gating/attention complexity; no compute numbers for most; two-stage training for SMU-Net [21], AdaMM [3]	Teacher/distillation hyperparameters unreported [3], [13], [19], [21]; no Pareto analysis of accuracy vs. parameters
Attention-Based Fusion	Chen et al. [1], Tongxue et al. [2]	Attention gates modulate fusion per modality availability [1]; single-modality WT 90.40% [1]	Modality identity for "Modality 1" unmapped [1]; qualitative missing-modality claims only [2]; no arbitrary subset evaluation	Availability-awareness limited to gating; no teacher election or dynamic routing
Adaptive Fusion	LCKD [13], SMML [18], M3AE [19]	Teacher election adapts to task [13]; hierarchical consistency (pixel+feature) + SAM priors [18]; masked autoencoder + model inversion + self-distillation [19]	LCKD: teacher selection heuristic, distillation temperature/scheduling NR [13]; SMML: SAM adaptation to MRI NR, dual-branch compute doubled, no BraTS edition specified [18]; M3AE: two-stage training, memory-efficient claim unverified [19]	No systematic comparison of teacher selection, distillation loss, temperature, architecture mismatch
Modality Dropout	mmFormer [9], M3AE [19], nnU-Net w/ dropout [11], [20]	Single model covers all subsets; mmFormer "outperforms SOTA on almost all subsets" [9]; M3AE "superior on BraTS 2018/2020" [19]	Requires all 15 non-empty subsets during training; subset sampling strategy NR; no worst-case subset reporting	No standardized 15-config benchmark; "all subsets" claims lack per-subset Dice/HD95 matrices
Modality-Specific Encoding	ShaSpec [7], Chen et al. [4], SMU-Net [21], mmFormer [9]	Separate encoders preserve modality-specific features; ShaSpec >3/5/3% gains [7]; Chen et al. >16% WT gain [4]	K encoders increase parameters; runs only available encoders at inference but all at training; distribution alignment complexity [7]	No ablation on encoder sharing vs. separation; compute overhead unquantified

Method Family	Representative Methods	Strengths (Evidence)	Limitations (Evidence/Gaps)	Unresolved Challenges
Shared Latent Representation (VAE)	HeMIS [23], [27]	Principled probabilistic embedding; handles arbitrary subsets naturally; similar to subset-specific nets [23]; FLAIR→WT, T1ce→TC/ET asymmetry confirmed [27]	VAE training instability; KL annealing NR; synthesis quality not primary goal; no missing-modality Dice matrices in context	Latent space dimensionality vs. robustness trade-off unexplored; no uncertainty from latent variance
Knowledge Distillation	LCKD [13], AdaMM [3], M3AE [19], SMU-Net [21]	Transfers full-modality knowledge to missing-modality paths; AdaMM avoids synthesis overhead [3]; LCKD teacher election adapts to task [13]; +3.61/5.99/3.76% Dice [13]; AdaMM +23.55% single-modality Dice vs generative [3]	Two-stage training (M3AE, SMU-Net); teacher selection heuristic (LCKD); distillation temperature/scheduling NR; AdaMM compute claim "lower overhead" unquantified [3]	No head-to-head on same BraTS edition with same missing configs; distillation vs. synthesis utility unresolved
Modality Synthesis/Reconstruction	FgC2F-UDiff [17], BraSyn methods [20], [22], MouseGAN++ [3], Hyper-GAE [3]	High synthesis quality (PSNR 25.96–27.93 dB, SSIM 0.881–0.919 on BraTS [17]); BraSyn-2025 SSIM% per modality [22]	**No downstream segmentation Dice reported in context**; synthesis adds generator compute; AdaMM beats generative methods on segmentation [3]	Pixel-level PSNR/SSIM ≠ segmentation utility; no joint benchmark of synthesis quality vs. downstream Dice
Adaptive Missing-Modality-Aware	AdaMM [3], SMML [18], LCKD [13], Chen et al. [1], [4]	Dynamic gating/attention/teacher selection based on available modalities; AdaMM GSME+LGRM [3]; SMML HCC+SAM [18]; LCKD teacher election [13]	Gating/selection mechanisms add complexity; SMML dual-branch compute; no inference latency numbers; availability mask consumption varies	No systematic evaluation of availability-awareness levels (Blind vs. Modality-Aware vs. Teacher Election vs. Dynamic Routing)

The table exposes a recurring pattern: methods claiming robustness to "all subsets" [9], [19] or "arbitrary missing modalities" [18], [23] rarely publish per-subset Dice/HD95 matrices, making cross-family comparison impossible. Conversely, methods with comprehensive 15-config evaluations [3] report only relative gains against selected baselines, not absolute Dice per configuration. This evidence gap directly motivates Gap 1 in Table 11 (standardized missing-modality benchmark).

6.2 Generalization, Robustness & Clinical/Deployment Concerns

Beyond segmentation accuracy, clinical deployment demands evidence across seven reliability dimensions. The retrieved passages provide the following sentence-level evidence counts (from EVIDENCE LIMITS), where a count of zero means **not observed in the retrieved passages**, not that no study has ever addressed it:

- **External / multi-centre validation**: 0 sentences. BraTS challenge test sets are multi-institutional but protocol-harmonized [11], [25], [26]; BraTS-SSA [29] highlights domain shift in Sub-Saharan Africa but no segmentation method evaluates on it.
- **Cross-validation**: 5 sentences (5-fold CV reported in [5], [6], [10], [16], [26]).
- **Patient-wise split**: 0 sentences. Splits are case-wise but not explicitly patient-wise (longitudinal independence unverified).
- **Uncertainty / calibration / OOD detection**: 2 sentences. EMMA [26] notes similar val/test performance across sources as a robustness proxy; no method reports predictive variance, entropy, ECE, Brier score, or failure detection metrics.
- **Random modality dropout during training**: 4 sentences ([9], [11], [19], [20] mention dropout or masking).
- **All modality subsets evaluated**: 0 sentences. Claims of "all subsets" [9], [19] or "arbitrary subsets" [18], [23] appear in prose but no 15/31/63-config matrix was extracted.
- **Code availability**: 4 of 31 sources mention code release.

Clinical reliability evidence void. No method in the corpus reports formal uncertainty quantification, calibration curves, or failure detection. External validation is limited to BraTS challenge hidden test sets (curated, same protocol). The BraSyn [20], [22] and METS [25] challenges are separate tasks, not evaluated by the segmentation methods surveyed here. Consequently, **the field has not yet evidenced** uncertainty-aware prediction, calibrated confidence, OOD/failure detection, or generalization to routine clinical distributions (different scanners, artifacts, post-operative cavities [12], low-quality MRI [29]).

Computational transparency gap. Only [8] reports epoch time (9 min/epoch, V100, 285 cases) and total training (2 days, 300 epochs). [1] report hardware (A100) and epochs (100) but not wall-clock time. Most papers omit FLOPs, parameters, inference latency, and memory footprint. AdaMM [3] claims "lower computational overhead" versus generative methods but provides no numbers. Without inference ms/case, GPU memory, and parameter counts, deployability in PACS, edge, or intra-operative settings cannot be assessed.

Robustness to absent/corrupted inputs. The consistent modality-contribution asymmetry (FLAIR→WT, T1ce→TC/ET, T1/T2 complementary) [2], [27] is the only robust generalization finding. Missing FLAIR causes sharp Dice degradation for WT; missing T1ce severely impairs TC/ET; missing T1/T2 yields milder drops [2]. This asymmetry informs teacher election [13], gating weights [1], [4], synthesis priority [20], [22], and evaluation design (single-modality FLAIR/T1ce missing are hardest cases). However, no method evaluates robustness to **corrupted** (motion, artifact) versus **absent** modalities, nor to partial volume effects in low-resolution clinical scans.

6.3 Limitations of this Review

This survey has four concrete limitations that bound its conclusions:

1. **Corpus-based narrative review, not exhaustive.** The curated internal repository contained 50 cataloged papers; 31 were retrieved by at least one hybrid query and form the evidence base. Nineteen catalog papers were not returned by any query and are therefore not discussed — a retrieval limitation, not an exclusion decision. Relevant work may have been missed.

2. **No independent re-run of published methods.** All quantitative claims are transcribed from original PDF passages. The Quantitative Evidence Ledger (Table 12) contains 217 verified numeric records from only 11 of the 31 cited papers; the other 20 contributed no verifiable record in the retrieved passages. The ledger is **not exhaustive** of every number in the corpus — it reflects only what the extractor could verify verbatim.

3. **Evaluation-protocol heterogeneity prevents meta-analysis.** As detailed in Section 5.1.1, BraTS edition mismatch (2012–2025), split inconsistency (5-fold CV, held-out validation, challenge test), missing-modality configuration non-standardization (predefined single/dual/triple vs. 15 combinatorial vs. arbitrary subsets), and metric scale variance (Dice 0–1 vs. %, HD95 mm, Sensitivity/Specificity ratios) make direct cross-study comparison impossible. This review therefore presents **qualitative synthesis with quantitative anchoring**, not a meta-analysis.

4. **Missing-modality degradation matrices not extractable.** No quantitative degradation matrix (per-modality, all 15 subsets, cross-study) could be built from the extracted records. Sources may discuss modality importance in prose (e.g., [2], [27]), but structured per-subset Dice/HD95 tables are absent from the retrieved passages. Consequently, claims such as "outperforms on all subsets" [9], [19] or "extensive experiments" [18] cannot be independently verified against a common benchmark.

These limitations mean that **RQ2** (methodological limitations preventing fair comparison) is answered by the review's own constraints as much as by the primary studies' heterogeneity. The proposed BraTS-Missing benchmark (Gap 1, Table 11) directly addresses Limitation 3 by fixing the 15/31/63 subset configurations, splits, hidden test, and metric suite.

6.4 Open Research Directions



Figure 8. Research roadmap linking evidence-supported gaps to concrete directions and evaluation protocols

Four concrete directions grounded in the evidence:

1. **Standardized missing-modality benchmark (from Gap 1, follows RQ2, RQ3).** Current evaluations use 15 configs [3], "all subsets" [9], [19], arbitrary subsets [18], [23], or single-modality dropout [1], [2], [7], [13] — no unified protocol exists. We propose **BraTS-Missing**: fixed 15/31/63 non-empty subset configurations, fixed train/val/test splits across BraTS editions, hidden test server, and mandatory reporting of Dice/HD95/Sensitivity/Specificity **plus** uncertainty metrics (predictive entropy, mutual information) per subset. This enables fair cross-family comparison and worst-case robustness analysis.
2. **Uncertainty quantification and calibration as mandatory reporting (from Gap 2, follows RQ3).** Zero methods in the corpus report calibration curves, ECE, Brier scores, or failure detection. Missing modalities inherently increase epistemic uncertainty; a model that cannot signal "I don't know" is unsafe for clinical triage. Future BraTS-Missing submissions should include an uncertainty head (MC dropout, deep ensembles, evidential learning, or conformal prediction) and report calibration on the held-out test set. The evaluation protocol must measure not only average Dice but also **calibration error stratified by missing-modality configuration**.
3. **External clinical validation beyond curated challenge sets (from Gap 3, follows RQ3).** BraTS challenge test sets are multi-institutional but protocol-harmonized [11], [25], [26]. BraTS-SSA [29] demonstrates substantial domain shift in low-resource settings, yet no segmentation method evaluates on it. We propose a **multi-center retrospective cohort track** with deliberate missing-modality patterns (contrast allergy, protocol omission, artifact corruption), reporting per-site Dice, worst-site Dice, and calibration shift. Domain generalization methods (e.g., test-time adaptation, invariant risk minimization) should be benchmarked on this track.
4. **Computational–clinical Pareto frontier reporting (from Gap 4, follows RQ2, RQ3).** Only [8] provides training time; inference latency, parameters, FLOPs, and GPU memory are absent for 30/31 methods. High-resource approaches (ensembles [8], Transformers [9], [10], diffusion [17]) may be impractical for PACS integration or intra-operative use. Every method should report **inference ms/case (batch=1), peak GPU memory, parameter count, and FLOPs** alongside Dice. Pareto plots (Dice vs. latency) will expose the deployability frontier and guide method selection for resource-constrained settings.

Additional directions (Gaps 5–7) address the synthesis-vs-distillation trade-off, teacher/distillation optimization, and failure mode taxonomy, each with concrete evaluation protocols in Figure 1.

7. Conclusion

RQ1. Across the four analytical dimensions, hybrid fusion with shared+specific encoders and adaptive missing-modality-aware strategies (teacher election, gated fusion, knowledge distillation) show the strongest evidence for robustness to missing modalities, but computational efficiency and clinical reliability remain unevidenced for all families. Early/late fusion lack missing-modality robustness; synthesis methods report image quality (PSNR/SSIM) without downstream segmentation Dice; no method reports uncertainty, calibration, or external clinical validation beyond BraTS challenge test sets.

RQ2. Methodological limitations preventing fair comparison and clinical translation are: (i) BraTS edition mismatch (2012–2025), (ii) split inconsistency (5-fold CV vs. held-out validation vs. hidden test), (iii) missing-modality configuration non-standardization (predefined vs. 15 combinatorial vs. arbitrary subsets), (iv) metric scale variance (Dice 0–1 vs. %, HD95 mm), (v) absent uncertainty/calibration/failure-detection reporting, and (vi) computational transparency gap (only 1/31 papers reports training time, 0/31 report inference latency/FLOPs/params).

RQ3. The seven evidence-supported gaps (Table 11) are all critical; the most immediately actionable are Gap 1 (standardized missing-modality benchmark) and Gap 2 (uncertainty quantification), as they enable rigorous evaluation of all other gaps. Gap 3 (external clinical validation) is essential for translation but requires multi-institutional coordination. Gaps 4–7 (computational Pareto, synthesis utility, distillation optimization, failure taxonomy) are tractable given the proposed BraTS-Missing infrastructure.

RQ4. The consistent modality-contribution asymmetry (FLAIR→WT, T1ce→TC/ET, T1/T2 complementary) [2], [27] should directly inform: (a) teacher election — select FLAIR teacher for WT, T1ce teacher for TC/ET [13]; (b) gating mechanisms — learn to upweight FLAIR for WT, T1ce for TC/ET [1], [4]; (c) synthesis priority — allocate generator capacity to FLAIR and T1ce [20], [22]; (d) evaluation protocols — mandate single-modality FLAIR-missing and T1ce-missing as hardest cases, and report per-subregion Dice per missing-modality configuration.

Outlook. The field has converged on hybrid encoder architectures with adaptive fusion and knowledge distillation as the dominant paradigm for missing-modality robustness. The next decade must shift from **accuracy-on-curated-benchmarks** to **reliability-in-the-wild**: standardized missing-modality benchmarks with uncertainty reporting, external validation on heterogeneous clinical cohorts, and computational–clinical Pareto transparency. The BraTS-Missing benchmark proposed here, if adopted by the community, would provide the common ground needed for the next generation of clinically deployable glioma segmentation systems.

Appendix A. Implementation and Evaluation Details

Datasets. BraTS editions 2012–2025 (adult glioma); BraTS-METS 2023 [25] and BraTS-PEDs 2023 [14] mentioned as separate tracks but not surveyed for fusion/missing-modality methods. BraSyn 2023/2025 [20], [22] provide synthesis benchmarks (219 glioma + 31 mets cases for BraSyn-2025 [22]). BraTS-SSA [29] highlights Sub-Saharan Africa domain shift.

Metrics. Dice (0–1 or %, preserved as reported), HD95 (mm), Sensitivity, Specificity (ratios), PSNR (dB), SSIM (0–1 or %). AURC (Area Under the Robustness Curve) used in [1] for missing-modality robustness across modality counts.

Missing-modality configurations. Predefined single-modality dropout [1], [2], [3], [7], [9], [13], [19], [21]; 15 combinatorial configs (single/dual/triple/full) [3]; arbitrary subsets [9], [18], [19], [23]; no standardized 31/63-config evaluation in corpus.

Evaluation splits. 5-fold cross-validation on training set [5], [6], [10], [16]; held-out validation [1], [8], [11], [26]; challenge hidden test [11], [25], [26]. Validation Dice consistently higher than test (e.g., [6]: Val WT 0.896 → Test WT 0.858).

Hyperparameters (reported). Chen et al. [1] employed a 3D U-Net backbone with shared and modality-specific encoders and early decoder fusion, optimized via Adam with a learning rate of 1e-2 and cosine annealing over 100 epochs with a batch size of 12 on an A100 GPU. Autoencoder Regularization [8] utilized an ensemble of 10 models trained for 300 epochs on a V100 32GB GPU, requiring 9 minutes per epoch (285 cases). nnU-Net [11] employs a self-configuring pipeline; no fixed hyperparameters are reported. FgC2F-UDiff [17] is a diffusion model; hardware not reported (NR). M3AE [19] adopts a two-stage approach (masked autoencoder, model inversion, and self-distillation), described as "memory-efficient"; hardware not reported (NR).

Code availability. Explicitly mentioned in 4/31 retrieved sources.

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